

PROBABILITY MEASURES ON PRODUCT SPACES

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1. Summary and Introduction

Let $X = \{x\}$ and $Y = \{y\}$ be two arbitrary sets and let A and B be two σ -fields of subsets of X and Y respectively. Let $Z = X \times Y$ and $C = A \times B$ be the corresponding product set and product σ -field respectively. Let μ be a probability measure on (Z, C) and let P and Q be the (marginal) probability measures induced by μ on (X, A) and (Y, B) respectively. Finally, let $\lambda = P \times Q$ be the product measure on (Z, C) corresponding to P and Q .

In Sections 3 and 4 we formalize and extend certain results of Lancaster (1958, 1959, 1963 where further references can be found) on the series expansion of the Randon-Ninodym derivative $\frac{d\mu}{d\lambda}$ and the concepts of canonical functions and correlations. In Section 5 some results on minimum mean square approximation are given and in Section 6 the special case of the normal distribution including the multivariate case is considered. Several open problems are also indicated.

2. Notation and Preliminaries

The terminology of Zaenen (1953) in connection with Hilbert space methods will be used.

We denote by $L_2(\mu)$ the space of all C -measurable and μ -square-integrable real valued functions on Z . Analogously for $L_2(\lambda)$, $L_2(P)$, $L_2(Q)$. Elements of $L_2(P)$ will usually be denoted by f with or without sub- or superscripts and elements of $L_2(Q)$ by g . The space $L_2(\mu)$ is a Hilbert space in the usual sense; inner product is defined by $(\phi_1, \phi_2)_\mu = \int \phi_1 \phi_2 d\mu$ and norm by $\|\phi\|_\mu = (\phi, \phi)_\mu^{1/2}$. Analogously for the Hilbert spaces $L_2(\lambda)$, $L_2(P)$ and $L_2(Q)$. We will drop the subscripts μ , λ , P , Q on the inner products and norms whenever this cannot give rise to confusion.

The restriction of attention to real valued functions is unnecessary and is made mostly because we have statistical applications in mind. Our results can be written down for complex valued functions in a straightforward manner.

It is convenient, although for many of our results not necessary, to assume that spaces $L_2(\mu)$, $L_2(\lambda)$, $L_2(P)$ and $L_2(Q)$ are separable in their norm topologies. We make this assumption throughout this paper. A sufficient condition for this to be the case is that the σ -fields A and B both be countably generated. This in turn will be the case if X and Y are separable metric spaces and A and B the σ -fields generated by the closed sets in their respective metric topologies.

Useful results can be obtained by exploiting a connection between this set-up and the theory of linear integral equations (see eg. Zaanen (1953), Chapters 13 - 17). To see what we mean here, suppose $\mu \ll \lambda$ and write $T(x,y) = \frac{d\mu}{d\lambda}$. Then $T(x,y)$ may be thought of as the kernel of a linear integral equation. Associated with $T(x,y)$ (and hence with μ) is the linear transformation T of $L_2(Q)$ into $L_2(P)$ defined by

$$Tg = \int T(x,y)g(y)dQ, \quad g \in L_2(Q) \quad (*)$$

and many of the properties of the kernel $T(x,y)$ (and hence of μ) is reflected in those of the transformation T . In particular, series expansions of $T(x,y)$, Tg , etc. in terms of the singular values and elements of T can be obtained readily (see Zaanen (1953), §6, page 491). These facts carry over to series expansion of $\frac{d\mu}{d\lambda}$, etc. Of course, the motivation for studying certain questions and the type of questions we are interested in here, are different from those of the theory of integral equations. For example, it is usual to take $L_2(P) \equiv L_2(Q)$ in the case of integral equations (this space being the starting point), but in our set-up $L_2(P) \equiv L_2(Q)$ amounts only to the rather special case $X \equiv Y$, $A \equiv B$ and μ on (Z,C) symmetric in the sense that $P \equiv Q$.

The definition (*) of the transformation T requires that $\mu \ll \lambda$. We can define T without this requirement as follows. For each $g \in L_2(Q)$ $(f,g)_\mu$, $f \in L_2(P)$, defines a bounded linear functional on $L_2(P)$. Hence, by the Riesz representation theorem there exists $Tg \in L_2(P)$ such that

$$(f, Tg)_P = (f, g)_\mu \quad \text{for all } f \in L_2(P), g \in L_2(Q). \quad (1)$$

It is easy to check that T thus defined is a linear transformation of $L_2(Q)$ into $L_2(P)$ and in case $\mu \ll \lambda$ it coincides with T defined by (*). We will henceforth use (1) as the definition of T . We remark that T is bounded and in fact has norm 1. Also, Tg may be thought of as the "conditional expectation of g given x ", for, what is meant by this phrase is the conditional expectation of g given the sub- σ -field $A \times Y$ of C which, by definition is an $A \times Y$ -measurable function (i.e. function of x only) h , say, satisfying the equation

$$\int_A h(x)dP = \int_{A \times Y} g(y)d\mu \quad (2)$$

for all $A \in \mathcal{A}$. Taking f the indicator function of A in (1) implies that $h = Tg$ satisfies (2).

We will refer to the transformation T defined by (1) as the transformation associated with μ .

3. Series Expansion of $\frac{d\mu}{d\lambda}$.

Theorem 1. Let $\{f_i, i=1,2,\dots\}$ and $\{g_i, i=1,2,\dots\}$ be complete orthonormal families in $L_2(P)$ and $L_2(Q)$ respectively (their existence follows from our separability assumption on $L_2(P)$ and $L_2(Q)$). Let

$$\rho_{ij} = (f_i, g_j)_\mu = \int f_i g_j d\mu. \quad (3)$$

(a) If $\Sigma \rho_{ij}^2 < \infty$, then $\mu \ll \lambda$ and

$$\int \left(\frac{d\mu}{d\lambda}\right)^2 d\lambda = \Sigma \rho_{ij}^2, \quad (4)$$

and

$$\frac{d\mu}{d\lambda} = \Sigma \rho_{ij} f_i g_j \quad \text{a.s. } \lambda \quad (5)$$

(convergence in $\|\cdot\|_\lambda$ -norm).

(b) If $\mu \ll \lambda$ and $\int \left(\frac{d\mu}{d\lambda}\right)^2 d\lambda < \infty$, then (4) and (5) hold.

Proof. First of all we remark that $\{f_i g_j\}$ is a complete orthonormal family in $L_2(\lambda)$. Now, in case (a) $\Sigma \rho_{ij}^2 < \infty$ and completeness of $\{f_i g_j\}$ imply that the family $\{\rho_{ij} f_i g_j\}$ is summable in $L_2(\lambda)$. Let

$$\phi = \Sigma \rho_{ij} f_i g_j \quad (6)$$

$$\text{with } \phi \in L_2(\lambda) \text{ and } \|\phi\|_\lambda^2 = \Sigma \rho_{ij}^2. \quad (7)$$

We need only show that

$$\mu(C) = \int_C \phi d\lambda \quad \text{for all } C \in \mathcal{C}, \quad (8)$$

and it will suffice to show this for sets C of the form $C = A \times B$ with $A \in \mathcal{A}$ and $B \in \mathcal{B}$. Denoting, in general, by I_D the indicator function of the set D , we have

$$\mu(A \times B) = \int I_A I_B d\mu = (I_A, I_B)_\mu. \quad (9)$$

But $I_A \in L_2(P)$ and $I_B \in L_2(Q)$. Hence substituting their Fourier expansions into (9), we have

$$\begin{aligned} \mu(A \times B) &= \sum_{i,j} \Sigma (I_A, f_i)(I_B, g_j)(f_i, g_j)_\mu \\ &= \sum_{i,j} \Sigma (I_A, f_i)(I_B, g_j) \rho_{ij} \\ &= \sum_{i,j} \Sigma \rho_{ij} (I_{A \times B}, f_i g_j)_\lambda \\ &= (I_{A \times B}, \phi)_\lambda \\ &= \int_{A \times B} \phi d\lambda, \end{aligned} \quad (10)$$

i.e. (8) holds, and part (a) follows. For part (b) we are

assuming that $\frac{d\mu}{d\lambda} \in L_2(\lambda)$. Its Fourier coefficients with respect to $\{f_i g_j\}$ are

$$\left(\frac{d\mu}{d\lambda}, f_i g_j\right)_\lambda = \int f_i g_j \frac{d\mu}{d\lambda} d\lambda = \int f_i g_j d\mu = \rho_{ij}. \quad (11)$$

(4) and (5) now follow from completeness of the family $\{f_i g_j\}$.

Remarks

(i) Part (b) of Theorem 1 was also proved by Lancaster (1958, 1963). He writes $\phi^2 + 1 = \int (\frac{d\mu}{d\lambda})^2 d\lambda$ and terms μ a ϕ^2 -bounded measure if $\phi^2 < \infty$. According to Theorem 1 this is equivalent to $\Sigma \rho_{ij}^2 < \infty$. In terms of the transformation T associated with μ , we have $\rho_{ij} = (f_i, Tg_j)_p$ and ϕ^2 -boundedness becomes equivalent to T being of finite double norm (a "Hilbert-Schmidt transformation"). Theorem 9 p245 of Zaenen (1953) has a content analogous to that of Theorem 1 here but in a somewhat different context.

(ii) ϕ^2 -boundedness is necessary and sufficient for the validity of the expansion (5) of $\frac{d\mu}{d\lambda}$ in the sense that the series in (5) converges in the $\|\cdot\|_\lambda$ -norm if and only if μ is ϕ^2 -bounded. The example given by Lancaster (1963) p536, 537 and the example in remark (i) following Theorem 2 below show that μ may be ϕ^2 -unbounded while an expansion of the form (5) for $\frac{d\mu}{d\lambda}$ still holds, the series in (5) now being a.s. λ convergent (actually absolutely in both examples). This raises the question of finding necessary and sufficient conditions on μ under which an expansion of the form (5) with a.s. λ convergence (or absolute a.s. λ convergence) of the series holds. Unfortunately no complete answer to this question seems to be available. Sufficient conditions can be found readily. Two approaches seem possible, viz.

(a) Using results in the theory of summation of Fourier series: Thus, if $\Sigma |\rho_{ij}| < \infty$, then it follows from the monotone convergence theorem and the Schwarz inequality that (5) holds, the series being absolutely a.s. λ convergent. Again, if for a particular order of summation $n(i,j)$ indicates the number of the (i,j) -th term and if $\Sigma \rho_{ij}^2 \log^2 n(i,j) < \infty$, then (5) holds, the series being a.s. convergent. This follows e.g. from Theorem B p458 of Loève (1960); see also Alexits (1961), Chapter II. These conditions are not necessary.

(b) The relation (10) may be written

$$\mu(A \times B) = \sum_{i,j} \rho_{ij} \int_{A \times B} f_i g_j d\lambda \quad (12)$$

and it follows that any condition guaranteeing the validity of interchanging integration and summation here will be a sufficient condition for the validity of (5) with a suitable sense of convergence. In the discrete case the situation becomes particularly simple, as is indicated in the following theorem.

Theorem 2. Let X and Y be subsets of the positive integers with A and B the corresponding power sets. Abbreviate $\mu(x,y) \equiv \mu(\{(x,y)\})$, $P(x) \equiv P(\{x\})$, $Q(y) \equiv Q(\{y\})$. It involves no loss of generality to assume that $P(x) > 0$ and $Q(y) > 0$ for all x and y . Let $\{f_i\}$ and $\{g_j\}$ be complete orthonormal families in $L_2(P)$ and $L_2(Q)$ respectively. Let

$$\rho_{ij} = \int f_i g_j d\mu = \int_{x,y} f_i(x) g_j(y) \mu(x,y). \quad (13)$$

Then

$$\frac{d\mu}{d\lambda} = \sum \rho_{ij} f_i g_j \quad (14)$$

the series converging everywhere. If $\rho_{ij} = 0$ for $i \neq j$, then the series also converges absolutely everywhere.

Proof. Taking $A = \{x\}$ and $B = \{y\}$ in (12) and noting that

$$\frac{d\mu}{d\lambda}(x,y) = \mu(x,y)/P(x)Q(y), \text{ the first part follows immediately.}$$

Further, let $\delta(x,u)=1$ for $x=u$ and 0 otherwise. Then, for each fixed u , $\delta(\cdot,u)/P(u) \in L_2(P)$ and has Fourier coefficient $f_i(u)$ so that

$$\sum_i |f_i(u)|^2 = \|\delta(\cdot,u)/P(u)\|^2 = 1/P(u). \quad (15)$$

Similarly

$$\sum_j |g_j(v)|^2 = 1/Q(v). \quad (16)$$

From Schwarz's inequality we get $|\rho_{ij}| \leq 1$ and applying it again, we find that if $\rho_{ij} = 0$ for $i \neq j$, then

$$\begin{aligned} \sum |\rho_{ii} f_i(x) g_i(y)| &\leq (\sum |f_i(x)|^2 \sum |g_i(y)|^2)^{\frac{1}{2}} \\ &= (1/P(x)Q(y))^{\frac{1}{2}} \\ &< \infty. \end{aligned} \quad (17)$$

Remarks.

(i) As an example take

$$f_i(x) = \delta(x,i)/\sqrt{P(x)}, \quad g_j(y) = \delta(y,i)/\sqrt{Q(y)}. \quad (18)$$

Then

$$\rho_{ij} = \frac{\mu(i,j)}{\sqrt{P(i)}\sqrt{Q(j)}}. \quad (19)$$

It is easy to choose μ in such a way that $\sum \rho_{ij}^2 = \infty$; e.g. $\mu(i,j) = 0$ for $i \neq j$ and > 0 for $i = j$. Then $P(i)=Q(i)=\mu(i,i)$ and $\rho_{ij} = \delta_{ij} = \delta(i,j)$.

(ii) To what extent the restriction to the discrete case in this theorem can be relaxed, is not known. It is clear that we need at least the condition $\mu \ll \lambda$ to be able even to talk about expansions of the form (5). No example showing that this condition is not in general sufficient for (5) with a.s. convergence of the series seems to exist.

(iii) Suppose μ is singular with respect to λ . Then there exists a set $D \in \mathcal{C}$ such that $\lambda(D) = 0$ while $\mu(CD^C) = 0$ for all $C \in \mathcal{C}$. Define $\frac{d\mu}{d\lambda} = \infty$ over D and 0 over D^C . One might expect that the series $\sum \rho_{ij} f_i g_j$ would be summable in some sense to $\frac{d\mu}{d\lambda}$ thus defined. Consider the following example. Let $X=Y=[0,\pi]$ with μ assigning its measure uniformly over the diagonal D connecting the points $(0,0)$ and (π,π) . Then P and Q are both uniform over $[0,\pi]$ and the families $\{1, \sqrt{2} \cos(ix)\}, \{1, \sqrt{2} \cos(jy)\}$ are complete orthonormal in $L_2(P)$ and $L_2(Q)$. We have $\rho_{ij} = \delta(i,j)$

and it is readily shown that while the series $1 + 2 \sum_i \cos(ix) \cos(iy)$ is divergent a.s. it is Abel summable to ∞ over D , except for the points $(0,0)$ and (π,π) , and to 0 over D^c . Another example of this type is the degenerate bivariate normal case. It would be interesting to establish the extent to which e.g. Abel summability of the series $\sum_{ij} \rho_{ij} f_i g_j$ could be related to $\frac{d\mu}{d\lambda}$ in general.

(iv) Further useful results follow from relation (12), the validity of which does not depend on any special assumptions on μ .

Corollary 1. μ is always completely determined by P, Q and $\{f_i\}$, $\{g_i\}$, $\{\rho_{ij}\}$.

Corollary 2. If $f_1 \equiv 1 = g_1$, then $\mu = \lambda$ (i.e. μ is an independent product measure) if and only if $\rho_{11} = 1$ and $\rho_{ij} = 0$ otherwise.

Corollary 3. $\mu = \lambda$ if and only if, for any $f \in L_2(P)$ and $g \in L_2(Q)$ we have $\text{Cov}(f, g) = 0$ where $\text{Cov}(f, g) = (f, g)_\mu - (f, 1)(g, 1)$, (i.e. f and g are uncorrelated).

Proofs. The first follows immediately from (12). For the second necessity follows from (3) and orthonormality while sufficiency follows from (12) which reduces to $\mu(A \times B) = \int_{A \times B} d\lambda = \lambda(A \times B)$. For the third necessity is trivial and sufficiency is immediate from Corollary 2.

Remarks. These corollaries are also proved by Lancaster (1963, 1959) for the case X and Y real spaces by somewhat longer and different arguments.

4. Canonical Functions and Correlations

Following Lancaster (1963) we now introduce the concepts of canonical functions (variables) and correlations.

Definition. If there exist orthonormal families $\{f_i^*\}$ and $\{g_i^*\}$ in $L_2(P)$ and $L_2(Q)$ respectively and a sequence of non-negative numbers $\{\rho_i^*\}$ such that $f_i^* = g_i^* = 1$ and $\rho_1^* = 1$ and for $i \geq 2$,

$$\rho_i^* = \sup \{ \int f g d\mu : \|f\|_P = \|g\|_Q = 1 \text{ and for } j = 1, 2, \dots, i-1 \}$$

$$(f, f_j^*)_P = 0 = (g, g_j^*)_Q, \quad (21)$$

$$= \int f_i^* g_i^* d\mu$$

then we call $\{f_i^*\}$ and $\{g_i^*\}$ canonical functions and $\{\rho_i^*\}$ corresponding canonical correlations of μ .

We now require necessary and sufficient conditions for the existence of canonical functions and correlations. Necessary conditions are not known at present but simple sufficient conditions are readily obtainable.

Theorem 3. If the transformation T associated with μ is compact

("completely continuous") then canonical functions and correlations exist. A sufficient condition for compactness of T is that it be of finite double norm, i.e. ϕ^2 -boundedness of μ .

Proof. The proof of the first part consists of a rather trivial generalization of the concepts of singular values and elements of a bounded linear transformation as defined and discussed by Zaanan (1953) in §18, p.432.

As to the second part, it is well known that T of finite double norm implies T compact (see eg. Zaanan (1953), Ex. D p.319 or Schatten (1960) Theorem 1 p.32).

Remarks

(i) In his discussion Lancaster (1963, §4, p.535) seems to imply that canonical functions always exist. That this is not so can be seen from the following example.

Example. Let $X = Y = \{1, 2, 3, \dots\}$ and A and B the power set of the integers. Using the abbreviations in Theorem 2, let

$$\mu(x, x) = 2^{-x}(1-a^x), \mu(x, x+1) = 2^{-x}a^x \quad (22)$$

for $x=1, 2, \dots$, and $\mu(x, y)=0$ for all other x and y . Here a is a constant with $0 < a < \frac{1}{2}$. Then $P(x)=2^{-x}$ for all x and

$$Q(1)=\frac{1}{2}(1-a) \text{ while } Q(y)=2^{-y}(1-a^y)+2^{-(y-1)}a^{y-1} \text{ for } y=2, 3, \dots$$

Now, the first canonical functions always exist, viz. $f_1^* = 1$, $g_1^* = 1$ and $\rho_1^* = 1$. Suppose that a second pair of canonical functions also exists, say $f \in L_2(P)$ and $g \in L_2(Q)$ with

$$(f, 1) = 0 = (g, 1) \text{ and } \|f\| = \|g\| = 1, \quad (23)$$

and let $\rho_2^* = (f, g)_\mu = \int f g d\mu$. Suppose that in fact $\rho_2^* = 1$; then

$\int [f - g]^2 d\mu = \int f^2 d\mu - 2 \int f g d\mu + \int g^2 d\mu = 0$. Hence $f = g$ a.s. μ , i.e. letting $J_x = \{y : \mu(x, y) > 0\}$ we must have $g(y) = f(x)$ if $y \in J_x$. Hence, if $y \in J_x \cap J_{x+1}$ we must have $f(x) = g(y) = f(x+1)$. By (22), for each x the sets J_x and J_{x+1} have a non-empty intersection, and it follows that both f and g must be constants, which contradicts (23). Hence we must have $\rho_2^* < 1$.

Now, let k be an integer larger than 2 and consider the functions $f_k \in L_2(P)$ and $g_k \in L_2(Q)$ defined by

$$f_k(x) = \begin{cases} \left[\frac{P(k+1)}{P(k)} \cdot \frac{1}{P(k) + P(k+1)} \right]^{\frac{1}{2}} & \text{for } x = k \\ - \left[\frac{P(k)}{P(k+1)} \cdot \frac{1}{P(k) + P(k+1)} \right]^{\frac{1}{2}} & \text{for } x = k+1 \\ 0 & \text{otherwise} \end{cases} \quad (24)$$

and with g_k similarly defined by replacing P with Q in (24). Then (23) is satisfied with $f = f_k$ and $g = g_k$. Straightforward but somewhat tedious calculations now show that we have $\int f_k g_k d\mu = 1 - ba^k$

as $k \rightarrow \infty$, where b is a positive constant. Hence we can choose k so that $1 - ba^k > \rho_2^*$. This contradicts the maximality of ρ_2^* as per its definition.

(ii) That the conditions of Theorem 3 are not necessary, is seen from the following example. Let X, Y be as in (i) above; let μ be such that $\mu(x, y) = 0$ if $x \neq y$ while $\mu(x, x) > 0$. Then $P = Q$, $L_2(P) = L_2(Q)$ and $Tg = g$ for all $g \in L_2(Q)$. Hence T is not compact. Nonetheless the canonical functions and correlations exist, viz. take $\{f_i^*\}$ any orthonormal system in $L_2(P)$ with $f_1^* \equiv 1$ and $g_i^* = f_i^*$, $\rho_i^* = 1$ for all i .

(iii) The next theorem, due to Lancaster (1963, 1958), is stated for reference purposes.

Theorem 4. Suppose canonical functions $\{f_i^\}$ and $\{g_i^*\}$ exist. If for $f \in L_2(P)$ we have $(f, f_k^*)_P = 0$ then also $(f, g_k^*)_\mu = 0$. Analogously if we interchange f, g and P, Q . In particular, $(f_i^*, g_j^*)_\mu = \rho_i^* \delta_{ij}$.*

Remark. The next results discuss series expansions of $\frac{d\mu}{d\lambda}$ in terms of the canonical functions and correlations.

Theorem 5

(a) If canonical functions and correlations $\{f_i^*\}$, $\{g_i^*\}$, $\{\rho_i^*\}$ exist and if $\Sigma \rho_i^{*2} < \infty$, then $\mu < \lambda$ and

$$\int \left(\frac{d\mu}{d\lambda}\right)^2 d\lambda = \Sigma \rho_i^{*2} \quad (25)$$

and

$$\frac{d\mu}{d\lambda} = \Sigma \rho_i^* f_i^* g_i^* \text{ a.s. } \lambda. \quad (26)$$

(convergence in l_1 -norm).

(b) If $\mu < \lambda$ and $\int \left(\frac{d\mu}{d\lambda}\right)^2 d\lambda < \infty$, then canonical functions and correlations exist by Theorem 3, and (25) and (26) hold.

(c) If there exist orthonormal functions $\{f_i^*\}$, $\{g_i^*\}$ in

$L_2(P)$, $L_2(Q)$ respectively with $f_1^* \equiv g_1^* \equiv 1$ and such that with $\rho_i^* = (f_i^*, g_i^*)_\mu$ we have $1 = \rho_1^* \geq \rho_2^* \geq \dots \geq 0$ (27)

and if $\mu < \lambda$ and (26) holds, then $\{f_i^*\}$, $\{g_i^*\}$, $\{\rho_i^*\}$ are canonical functions and correlations, and (25) holds.

Remark. This theorem could be proved by extending the methods used in the proofs of Theorems 3, 4, 5 and 6 on pages 392, 393

of Zaanen (1953). Instead of doing this we deduce it from Theorems 1 and 4 above.

Proof. For (a) and (b), if $\{f_i^*\}$ is not complete in $L_2(P)$, let $\{e_k\}$ be a complete orthonormal system in the orthogonal complement of the span of $\{f_i^*\}$. Then $\{f_i^*\} \cup \{e_k\}$ is a complete orthonormal family in $L_2(P)$. Similarly, let $\{g_j^*\} \cup \{h_k\}$ be a complete orthonormal family in $L_2(Q)$. By Theorem 4 we have

$$(f_i^*, g_j^*)_\mu = \rho_i^* \delta_{ij} \text{ and } (f_i^*, h_k)_\mu = (e_k, g_j^*)_\mu = 0, \quad (28)$$

for all i, j and k . Further $\int \left(\frac{d\mu}{d\lambda}\right)^2 d\lambda < \infty$ implies, according to

Theorem 1, that $\Sigma \rho_i^{*2} < \infty$; hence in both cases $\rho_i^* \rightarrow 0$ as $i \rightarrow \infty$.

According to the maximality of ρ_i^* , $|(e_k, h_m)_\mu| \leq \rho_i^*$ for all i .

$$\text{Thus } (e_k, h_m)_\mu = 0 \quad (29)$$

for all k, m . Hence parts (a) and (b) of this Theorem follow from a direct application of parts (a) and (b) of Theorem 1.

For part (c), let $f \in L_2(P)$ and $g \in L_2(Q)$ with $\|f\| = \|g\| = 1$

and $(f, f_1^*) = 0 = (g, g_1^*)$. Then, using (26), $(f, g)_\mu = \int f g d\mu$

$= (fg, \frac{d\mu}{d\lambda})_\lambda = \Sigma \rho_i^* (fg, f_i^* g_i^*)_\lambda = \Sigma_{i \geq 2} \rho_i \alpha_i \beta_i$ where $\{\alpha_i\}, \{\beta_i\}$ are the Fourier coefficients of f and g respectively with respect to $\{f_i^*\}, \{g_i^*\}$. By the Schwarz inequality

$$(f, g)_\mu \leq \left(\Sigma_{i \geq 2} \rho_i \alpha_i^2\right)^{\frac{1}{2}} \left(\Sigma_{i \geq 2} \rho_i \beta_i^2\right)^{\frac{1}{2}} \quad (30)$$

and subject to the conditions $\Sigma \alpha_i^2 \leq 1$, $\Sigma \beta_i^2 \leq 1$ implied by $\|f\| = \|g\| = 1$, the righthand side of (30) has maximum value

$\sqrt{\rho_1} \sqrt{\rho_1} = \rho_1$ due to (27). The choice $\alpha_2 = 1 = \beta_2$, $\alpha_i = \beta_i = 0$ for $i > 2$, i.e. $f = f_2$, $g = g_2$, yields this maximum value for $(f, g)_\mu$. Repeating this argument, part (c) follows.

Remarks

(i) Parts (b) and (c) of this Theorem are also proved by Lancaster (1963, 1958) by essentially the same methods.

(ii) The discussion following Theorem 1 carries over. Thus, if $\Sigma \rho_i^{*2} < \infty$ ($\Sigma \rho_i^{*2} \log_i^2 < \infty$) then (26) holds the series being absolutely a.s. convergent (a.s. convergent). These conditions are not necessary. The equivalent of Theorem 2 follows.

Theorem 6. In the set-up of Theorem 2, if canonical functions and correlations $\{f_i^*\}, \{g_i^*\}$ and $\{\rho_i^*\}$ exist and if either $\rho_i^* \rightarrow 0$ as

$i \rightarrow \infty$ or at least one of the families $\{f_i^*\}$ or $\{g_i^*\}$ are complete,

$$\text{then } \frac{d\mu}{d\lambda} = \sum \rho_i^* f_i^* g_i^* \quad (31)$$

the series being absolutely convergent everywhere.

Proof. Under the conditions of the theorem (28) and (29) continue to hold and the theorem follows from Theorem 2.

Definition. The second canonical correlation ρ_2^* is called the maximal correlation of the product measure μ .

Corollary 4. $\mu = \lambda$ if and only if $\rho_2^* = 0$

Proof. Necessity is trivial and sufficiency follows from Corollary 2 or 3 and the definition of ρ_2^* .

Remarks. The notion of maximal correlation was introduced by Gebelein (1941) for the case X and Y real spaces, by Sarmanov and Zaharov (1960) for the case X and Y subsets of finite dimensional euclidian space. The corollary is also proved by Lancaster (1959) for the case X and Y real spaces and by Sarmanov (1958) and Sarmanov and Zaharov (1960) for the case X and Y subsets of finite dimensional euclidian space and μ ϕ^2 -bounded.

5. Approximation Problems

Some further properties of the transformation T are discussed here.

Theorem 7. Let $g \in L_2(Q)$ be fixed. Then the choice $f = Tg$ minimizes the distance $\|f - g\|_\mu$ and maximizes the correlation $(f, g)_\mu / \|f\| \|g\|$ over all $f \in L_2(P)$. In these senses Tg is the best approximator of g among all f in $L_2(P)$.

Proof. Tg being the conditional expectation of g given x (see §2) this result is really well known. Even so we give a proof because it appeals only to the definition of T . From (1) and Schwarz's inequality

$$(f, g)_\mu / \|f\| \|g\| = (f, Tg)_P / \|f\| \|g\| \leq \|Tg\| / \|g\| = \rho_g,$$

say, and equality is achieved by taking $f = Tg$. Also, using this result,

$$\begin{aligned} \|f - g\|_\mu^2 &= \|f\|^2 - 2(f, g)_\mu + \|g\|^2 \\ &\geq \|f\|^2 - 2\|f\| \|Tg\| + \|g\|^2 \\ &= (\|f\| - \|Tg\|)^2 + \|g\|^2 - \|Tg\|^2 \\ &\geq \|g\|^2 - \|Tg\|^2 \\ &= \|g\|^2 (1 - \rho_g^2) \end{aligned} \quad (32)$$

and equality is achieved for $f = Tg$. Note the familiar form of

$$\min \|f - g\|_\mu^2.$$

Remarks

(i) Interchanging f, g and P, Q and T, T' (adjoint of T) the best approximator of a given $f \in L_2(P)$ is $T'f$ in $L_2(Q)$.

(ii) With $\{f_i\}$ and $\{g_i\}$ and $\{\rho_{ij}\}$ as in Theorem 1 we have the expansion $Tg = \sum (f_i, Tg)_P f_i$

$$\begin{aligned} &= \sum (f_i, g)_\mu f_i \\ &= \sum_i \sum_j (g, g_j) \rho_{ij} f_i \end{aligned} \quad (33)$$

on substituting for g its Fourier expansion in terms of $\{g_j\}$.

If canonical functions and correlations exist this expansion can be written in terms of them.

Theorem 8. If canonical functions and correlations $\{f_i^*\}$, $\{g_i^*\}$ and $\{\rho_i^*\}$ exist and if either $\rho_i^* \rightarrow 0$ as $i \rightarrow \infty$ or at least one of the systems $\{f_i^*\}$ or $\{g_i^*\}$ is complete, then

$$Tg = \sum \rho_i^* (g, g_i^*) f_i^* \quad (34)$$

the series converging in $\|\cdot\|_P$ -norm. If μ is ϕ^2 -bounded this

series also converges absolutely a.s. P. Analogous results hold for $T'f$.

Proof. Choosing for $\{f_i\}$ a system including $\{f_i^*\}$ and similarly

for $\{g_j\}$, as in the proof of Theorem 5, (28) and (29) continue to hold and (34) follows from (33). Also, by the Schwarz and Bessel inequalities,

$$|\rho_i^* (g, g_i^*) f_i^*| \leq (\sum |(g, g_i^*)|^2)^{\frac{1}{2}} (\sum \rho_i^{*2} f_i^{*2})^{\frac{1}{2}} \leq \|g\| (\sum \rho_i^{*2} f_i^{*2})^{\frac{1}{2}}$$

and since

$$\int \sum \rho_i^{*2} f_i^{*2} dP = \sum \rho_i^{*2} \int f_i^{*2} dP = \sum \rho_i^{*2} < \infty$$

under ϕ^2 -boundedness, the rest of the theorem follows.

Remarks

(i) This theorem is related to Theorems 3 and 5 page 492, 493 of Zaanen (1953).

(ii) Under the conditions of this theorem, one can write ρ_g and

$$\begin{aligned} \min_f \|f - g\|_\mu^2 &\text{ in various forms, e.g. from (32) and (34),} \\ \min_{f \in L_2(P)} \|f - g\|_\mu^2 &= \|g\|^2 - \sum \rho_i^{*2} (g, g_i^*)^2. \end{aligned} \quad (35)$$

(iii) For the finite case (X, Y finite sets) finding the f having maximum correlation with a given g is briefly discussed as a problem due to Yates by Lancaster (1958) where (34) for this case

is also given. This problem has been discussed widely in the literature in various forms.

6. The Normal Case; Hermite Polynomials

6.1 Hermite polynomials of a real variable

For $x \in \mathbb{R}$, the real line, we define

$$h_r(x) = (-1)^r e^{\frac{1}{2}x^2} \left(\frac{d}{dx}\right)^r e^{-\frac{1}{2}x^2}, \quad r = 0, 1, \dots \quad (36)$$

and call h_r the r -th Hermite polynomial. The following are well-known facts in connection with these polynomials.

(i) Denoting by Q the probability measure on $\mathbb{R}$ corresponding to the unit normal distribution, i.e. with dx denoting Lebesgue measure, $\frac{dQ}{dx} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$, the family $\{\frac{1}{\sqrt{r!}} h_r(x)\}$ is a complete orthonormal system in $L_2(Q)$.

$$(ii) \int_{-\infty}^{\infty} e^{-itx} h_r(x) dQ = (\underline{i} t)^r e^{-\frac{1}{2}t^2} \quad (37)$$

for all real t , where $\underline{i}$ denotes the unit complex number, i.e.

$\underline{i} = \sqrt{-1}$. We will use this notation throughout this section.

$$(iii) \int_{-\infty}^{\infty} (\underline{i} t)^r e^{-\frac{1}{2}t^2} e^{-itx} dt = h_r(x) \frac{dQ}{dx} \quad (38)$$

for all real x .

6.2 Hermite polynomials of several variables

The following notation will be used for the remainder of this section. $\mathbb{R}^p$ denotes p -dimensional euclidian space; an element of $\mathbb{R}^p$ is thought of as a column vector; for $x \in \mathbb{R}^p$, x' denotes the transpose of the vector x ; thus we have $x' = (x_1, \dots, x_p)$ where x_i , $i = 1, \dots, p$ denote the components of x . We write

$$\nabla(x)' = \left(\frac{\partial}{\partial x_1}, \dots, \frac{\partial}{\partial x_p} \right). \quad (39)$$

Now, let $\dagger$ be a positive definite symmetric matrix of order p and let $\alpha \in \mathbb{R}^p$. We define

$$H_r(x; \dagger, \alpha) = (-1)^r e^{\frac{1}{2}x'\dagger^{-1}x} (\alpha' \nabla(x))^r e^{-\frac{1}{2}x'\dagger^{-1}x} \quad (40)$$

and call $H_r(x; \dagger, \alpha)$ the r -th general Hermite polynomial with parameters $\dagger$ and α .

Hermite polynomials of several variables have been defined and studied fairly extensively in the literature, see e.g. Erdélyi (1953). The definition (40) seems to allow greater generality than the existing definitions due to the parameter α introduced here. We will now develop the properties of these polynomials that will be needed here.

First we note that if U is a $p \times p$ non-singular matrix, then we have $\nabla(Ux) = U'^{-1} \nabla(x)$ (41)
where U' is the transpose of U .

In particular, choosing U such that $U' \mathbb{I}^{-1} U = I$, (42)
where I is the $p \times p$ identity matrix, we get

$$H_r(Ux; \mathbb{I}, \alpha) = H_r(x; I, U^{-1}\alpha). \quad (43)$$

We will now establish a connection between h_r and H_r . Writing,

for the norm of a vector $\alpha \in R^p$, $\|\alpha\| = (\Sigma \alpha_i^2)^{\frac{1}{2}}$, suppose that
 $\|\alpha\| = 1$ and let V be a $p \times p$ orthogonal matrix with first column
 α . Then $\alpha'V = (1, 0, \dots, 0)$ and $V(Vy) = VV(y)$. (44)

$$\begin{aligned} \text{Hence } H_r(Vy; I, \alpha) &= (-1)^r e^{\frac{1}{2}\|y\|^2} [\alpha'VV(y)]^r e^{-\frac{1}{2}\|y\|^2} \\ &= (-1)^r e^{\frac{1}{2}\|y\|^2} \left(\frac{\partial}{\partial y_1}\right)^r e^{-\frac{1}{2}\|y\|^2} \\ &= h_r(y_1) \end{aligned} \quad (45)$$

in view of (36). Setting $x = Vy$ so that $y = V'x$ and
 $y_1 = \alpha'x$, we have

$$H_r(x; I, \alpha) = h_r(\alpha'x) \quad (46)$$

for all x , where $\|\alpha\| = 1$. From the definition, we have

$$H_r(x; I, c\alpha) = c^r H_r(x; I, \alpha) \quad (47)$$

for a real constant c . Hence, for an arbitrary $\alpha \in R^p$, $\|\alpha\| \neq 0$,

$$H_r(x; I, \alpha) = \|\alpha\|^r h_r\left(\frac{\alpha'x}{\|\alpha\|}\right). \quad (48)$$

Choosing U as in (42) and writing $x = Uy$, we have, by (43)

$$H_r(x; \mathbb{I}, \alpha) = H_r(Uy; \mathbb{I}, \alpha) = H_r(y; I, U^{-1}\alpha)$$

$$= \|U^{-1}\alpha\|^r h_r\left[\frac{\alpha'U^{-1}x}{\|U^{-1}\alpha\|}\right], \text{ by (48).}$$

$$\text{But, by (42) } \mathbb{I}^{-1} = U'^{-1}U^{-1} \quad (49)$$

so that $\|U^{-1}\alpha\|^2 = (U^{-1}\alpha)'(U^{-1}\alpha) = \alpha'\Sigma^{-1}\alpha$ and $\alpha'U'^{-1}y =$

$\alpha'U'^{-1}U^{-1}x = \alpha'\mathbb{I}^{-1}x$. Substituting these facts we get

$$H_r(x; \mathbb{I}, \alpha) = (\alpha'\mathbb{I}^{-1}\alpha)^{r/2} h_r\left[\frac{\alpha'\mathbb{I}^{-1}x}{|\alpha'\mathbb{I}^{-1}\alpha|^{\frac{1}{2}}}\right]. \quad (50)$$

Next we derive a generalization of (37). Suppose $\|\alpha\|=1$, $t \in R^p$
and consider the integral

$$\int_{R^p} H_r(x; I, \alpha) e^{it'x} \left(\frac{1}{2\pi}\right)^{p/2} e^{-\frac{1}{2}\|x\|^2} dx, \quad (51)$$

where dx denotes integration w.r.t. Lebesgue measure in R^p .
The transformation $x = Vy$ with V as defined above, together
with (45) and (37) are readily seen to reduce this integral to

$$(\frac{1}{2\pi} \alpha't)^r e^{-\frac{1}{2}\|t\|^2}. \quad (52)$$

Now denoting by P the probability measure in R^P corresponding to the density with respect to Lebesgue measure given by

$$\frac{dP}{dx} = \left(\frac{1}{2\pi}\right)^{P/2} \frac{1}{|\mathbb{A}|^{1/2}} e^{-\frac{1}{2}x'\mathbb{A}^{-1}x} \quad \text{where } |\mathbb{A}| \text{ is the determinant of } \mathbb{A},$$

and using the transformation $x = Uy$ with U as given by (42), it is readily seen that

$$\int_{R^P} H_r(x; \mathbb{A}, \alpha) e^{-\frac{1}{2}t'x} dP = (\underline{\alpha}'t)^r e^{-\frac{1}{2}t'\mathbb{A}t} \quad (53)$$

which is the required generalization of (37). Inverting (53),

$$\text{we have } \int_{R^P} (\underline{\alpha}'t)^r e^{-\frac{1}{2}t'\mathbb{A}t} e^{-\frac{1}{2}t'x} dt = H_r(x; \mathbb{A}, \alpha) \frac{dP}{dx} \quad (54)$$

which is the generalization of (38). Next we consider the orthogonality properties of the polynomials H_r . It follows directly from the definition that

$$H_r(x; I, \alpha) = \sum_{\substack{k_1, \dots, k_p \geq 0 \\ k_1 + \dots + k_p = r}} \frac{r!}{k_1! \dots k_p!} \prod_{i=1}^p \alpha_i^{k_i} h_{k_i}(x_i). \quad (55)$$

This result together with the orthogonality properties of $\{h_r\}$ readily show that

$$\int_{R^P} H_r(x; I, \alpha) H_s(x; I, \beta) \left(\frac{1}{2\pi}\right)^{P/2} e^{-\frac{1}{2}\|x\|^2} dx = (\alpha'\beta)^r r! \delta_{rs} \quad (56)$$

for $\alpha, \beta \in R^P$. Using the transformation $x = Uy$ with U as in (39) together with (40) and (56) it follows that

$$\int_{R^P} H_r(x; \mathbb{A}, \alpha) H_s(x; \mathbb{A}, \beta) dP = (\alpha'\mathbb{A}^{-1}\beta)^r r! \delta_{rs}, \quad (57)$$

from which it follows that

$$\left\{ \frac{1}{\sqrt{r!}} (\alpha'\mathbb{A}^{-1}\alpha)^{-r/2} H_r(x; \mathbb{A}, \alpha) \right\} \quad (58)$$

is an orthonormal family in $L_2(P)$. Also, if $\alpha^{(1)}, \dots, \alpha^{(p)}$ are

$$\text{such that } \alpha^{(i)'} \mathbb{A}^{-1} \alpha^{(j)} = 0 \quad \text{for } i \neq j, \text{ then} \quad (59)$$

$$\left\{ \frac{1}{\sqrt{r!}} (\alpha^{(i)'} \mathbb{A}^{-1} \alpha^{(i)})^{-r/2} H_r(x; \mathbb{A}, \alpha^{(i)}) \right\} \quad (60)$$

is an orthonormal family for each fixed r .

6.3 The real case

Let $X \equiv Y \equiv R$ with A and B the Borel sets of the real line. Then Z is the real plane and C the Borel sets of the plane. Let μ correspond to the unit bivariate normal distribution with correlation coefficient ρ , where $|\rho| < 1$. Then P and Q correspond

to the unit normal distribution on R and we have the well-known Mehler identity (see Erdélyi, 1953),

$$\frac{1}{\sqrt{1-\rho^2}} e^{-\frac{1}{1-\rho^2} \{\rho^2(x^2+y^2)-2\rho xy\}} = \sum_{r=0}^{\infty} \rho^r \frac{1}{r!} h_r(x) h_r(y) \quad (61)$$

The left-hand side is exactly $\frac{d\mu}{d\lambda}$ and it follows from Theorem 5(c) that we can take canonical functions to be $\{\frac{1}{\sqrt{r!}} h_r(x)\}$ and $\{\frac{1}{\sqrt{r!}} \text{sgn}(\rho) h_r(y)\}$ and the canonical correlations are $\{|\rho|^r\}$.

The series in (61) converges both in mean square and absolutely almost everywhere (Lebesgue). We have

$$\int \left(\frac{d\mu}{d\lambda}\right)^2 d\lambda = \frac{1}{1-\rho^2}. \quad (62)$$

In the case $\rho = 1$ μ is singular with respect to λ ; it concentrates its measure on the diagonal $D = \{(x, x) ; x \in R\}$ which has zero measure under λ . In this case the series in (60) is still Abel summable a.s. to $\frac{d\mu}{d\lambda}$ as given in Remark (iii) following Theorem 2.

6.4 The multiple correlation case

Let $X_j \in R^p$, $Y \in R$ so that $Z \in R^{p+1}$, with A, B, C corresponding Borel sets. We will use the notation of Section 6.2 here. Let μ correspond to the non-degenerate $(p+1)$ -dimensional normal distribution with mean vector zero and covariance matrix

$$\Gamma = \begin{bmatrix} \ddagger & \gamma \\ \gamma & \sigma^2 \end{bmatrix}$$

where $\ddagger$ is a $p \times p$ positive definite symmetric matrix, γ is a p -dimensional vector and $\sigma^2 > 0$. Then P corresponds to the p -dimensional normal distribution with mean zero and covariance matrix $\ddagger$ and Q corresponds to the univariate normal with mean zero and variance σ^2 .

Writing $v = \begin{pmatrix} t \\ u \end{pmatrix}$ with $t \in R^p$, $u \in R$, the characteristic function

$$\begin{aligned} \text{of } \mu \text{ is } \int e^{i v' z} d\mu &= e^{-\frac{1}{2} v' \Gamma v} \\ &= e^{-\frac{1}{2} t' \ddagger t} e^{-\frac{1}{2} \sigma^2 u^2} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{r!} (\sigma u)^r \left(\frac{t' \gamma}{\sigma} \right)^r \right]. \end{aligned}$$

Multiplying by $e^{-i v' z} = e^{-i t' x} e^{-i u y}$ and noting that the series is absolutely convergent, the absolute series converging to an integrable function so that we can integrate term-by-term, it

follows that $\frac{d\mu}{dz} = \int_{R^{p+1}} e^{-\frac{1}{2} v' \Gamma v} e^{-i v' z} dv$

$$= \sum_{r=0}^{\infty} \frac{1}{r!} \left[\int_{R^p} (\underline{i}\sigma u)^r e^{-\frac{1}{2}\sigma^2 u^2} e^{-\underline{i}uy} \right]_x \\ \left[\int_{R^p} \left(\underline{i} \frac{t'Y}{\sigma} \right)^r e^{-\frac{1}{2}t't} e^{-\underline{i}t'x} dt \right].$$

Using (38) and (57) this becomes

$$\frac{d\mu}{dz} = \frac{dP}{dx} \cdot \frac{dQ}{dy} \cdot \sum_{r=0}^{\infty} \frac{1}{r!} h_r \left(\frac{Y}{\sigma} \right) H_r \left(x ; \frac{1}{\sigma} \right)$$

i.e.

$$\frac{d\mu}{d\lambda} = \sum_{r=0}^{\infty} \left(\frac{Y' \frac{1}{\sigma} Y}{\sigma^2} \right)^{r/2} \frac{1}{\sqrt{r!}} h_r \left(\frac{Y}{\sigma} \right) \frac{1}{\sqrt{r!}} \left(\frac{Y' \frac{1}{\sigma} Y}{\sigma^2} \right)^{-r/2} H_r \left(x ; \frac{1}{\sigma} \right)$$

and it follows from the orthogonality properties of the Hermite polynomials and Theorem 5(c) that canonical functions are

$$\left\{ \frac{1}{\sqrt{r!}} h_r \left(\frac{Y}{\sigma} \right) \right\}, \left\{ \frac{1}{\sqrt{r!}} \left(\frac{Y' \frac{1}{\sigma} Y}{\sigma^2} \right)^{-r/2} H_r \left(x ; \frac{1}{\sigma} \right) \right\}$$

and the canonical correlations are

$$\left\{ \left(\frac{Y' \frac{1}{\sigma} Y}{\sigma^2} \right)^{r/2} \right\}.$$

We note that the second canonical correlation ($r = 1$) is exactly the ordinary multiple correlation coefficient, (Anderson (1958)) between the measures P and Q . The corresponding canonical functions are respectively $\frac{Y}{\sigma}$ and $\frac{Y' \frac{1}{\sigma} x}{\sigma}$. It follows that the best μ -mean square approximator in $L_2(P)$ (i.e. among both linear and

non-linear functions on R^p) to y is $Y' \frac{1}{\sigma} x$.

6.5 The multivariate case.

Let $X \equiv R^p$, $Y \equiv R^q$ so that $Z \equiv R^{p+q}$. We will suppose that $p \geq q$. Let μ correspond to the multivariate normal distribution on Z with mean zero and covariance matrix

$$\Gamma = \begin{bmatrix} \dagger_{11} & \dagger_{12} \\ \dagger_{21} & \dagger_{22} \end{bmatrix}$$

where $\dagger_{11}$ is $p \times p$ positive definite, $\dagger_{22}$ is $q \times q$ positive definite and $\dagger_{12} = \dagger_{21}'$ is $p \times q$. Then P and Q correspond to multivariate normal distributions on X and Y with zero means and covariance matrices $\dagger_{11}$ and $\dagger_{22}$ respectively.

Writing $v = \begin{pmatrix} t \\ u \end{pmatrix}$ with $t \in R^p$, $u \in R^q$ the characteristic function of μ is $\int e^{-\underline{i}v'z} d\mu = e^{-\frac{1}{2}v'\Gamma v}$

$$= e^{-\frac{1}{2}t' \dagger_{11} t} e^{-\frac{1}{2}u' \dagger_{22} u} \left[\sum_{r=0}^{\infty} \frac{(-1)^r}{r!} (u' \dagger_{21} t)^r \right]. \quad (63)$$

Now, let $\rho_1, \dots, \rho_q$ be the ordinary canonical correlations and $\{\alpha^{(1)}, \dots, \alpha^{(q)}\}, \{\beta^{(1)}, \dots, \beta^{(q)}\}$ the corresponding coefficients of canonical variates (see Anderson (1958), Hotelling (1936)). These quantities are characterized by the relations:

$$\left. \begin{aligned} 1 \geq \rho_1 \geq \dots \geq \rho_q \geq 0 \\ \alpha^{(i)} \in R^p, \quad \beta^{(i)} \in R^q \\ \text{and} \\ \alpha^{(i)'} \dagger_{11} \alpha^{(i)} = \delta_{ij} = \beta^{(i)'} \dagger_{22} \beta^{(j)} \\ \text{and} \\ \alpha^{(i)'} \dagger_{12} \beta^{(j)} = \rho_i \delta_{ij}. \end{aligned} \right\} \quad (64)$$

It is easily seen that for $t \in R^p$ and $u \in R^q$ we have

$$\left. \begin{aligned} t &= \sum_1^p (t' \dagger_{11} \alpha^{(i)}) \alpha^{(i)} \\ u &= \sum_1^q (u' \dagger_{22} \beta^{(i)}) \beta^{(i)} \end{aligned} \right\} \quad (65)$$

Hence,

$$u' \dagger_{21} t = \sum_1^q \rho_i (u' \dagger_{22} \beta^{(i)}) (t' \dagger_{11} \alpha^{(i)}) \quad (66)$$

where we have used (64). Substituting (66) into (63), multiplying both sides by $e^{-\frac{1}{2}v't} = e^{-\frac{1}{2}t'x} e^{-\frac{1}{2}u'y}$ and integrating term-by-term with respect to t , we get after simplification,

$$\frac{d\mu}{d\lambda} = \int_0^\infty \prod_{i=1}^q \rho_i \frac{1}{\sqrt{k_i!}} H_{k_i}(x; \dagger_{11}, \alpha^{(i)}) H_{k_i}(y; \dagger_{22}, \beta^{(i)}) \quad (67)$$

In view of the orthogonality relations for the Hermite polynomials and Theorem 5(c), canonical functions are

$$\left\{ \prod_{i=1}^q \frac{1}{\sqrt{k_i!}} H_{k_i}(x; \dagger_{11}, \alpha^{(i)}) \right\}, \left\{ \prod_{i=1}^q \frac{1}{\sqrt{k_i!}} H_{k_i}(y; \dagger_{22}, \beta^{(i)}) \right\} \quad (68)$$

and the canonical correlations are

$$\left\{ \prod_{i=1}^q \rho_i \right\}. \quad (69)$$

We note that the ordinary canonical functions (variates) appear among these as those linear in the components of x and y . They have maximum correlations with each other (not only among linear functions of these components). This was, according to Sarmanov and Zaharov (1960) already pointed out by Kolmogorov.

7. Acknowledgement

The author is indebted to the referee for helpful suggestions. The referee pointed out that an example to show that the canonical correlations might not always be attained was also given by Csaki and Fischer (1960) Magyar Tud. Akad. Mat. Kutato Int. Kozl. 5: 311.

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